

terminals	::=	
		$\rightarrow$
		$\rightarrow$
		$\Rightarrow$
		λ
		$\mapsto$
		$\vdash$
		$\emptyset$
		$\times$
		$<:$
		$\langle$
		$\rangle$
formula	::=	judgement
t	::=	terms: 0 succ t pred t iszero t (t) true false if t₁ then t₂ else t₃ constant zero successor predecessor zero test M constant true constant false conditional
v	::=	values: 0 succ v true false zero value successor value true value false value
Jop	::=	t $\rightarrow$ t' Evaluation
judgement	::=	Jop
user_syntax	::=	terminals formula t v

$t \rightarrow t'$ *Evaluation*

$$\frac{t_1 \rightarrow t'_1}{\text{succ } t_1 \rightarrow \text{succ } t'_1} \quad \text{E_SUCC}$$

$$\frac{}{\text{pred } 0 \rightarrow 0} \quad \text{E_PREDZERO}$$

$$\begin{array}{c}
\frac{}{\text{pred}(\text{succ } v_1) \longrightarrow v_1} \quad \text{E_PRED_SUCC} \\
\\
\frac{t_1 \longrightarrow t'_1}{\text{pred } t_1 \longrightarrow \text{pred } t'_1} \quad \text{E_PRED} \\
\\
\frac{}{\text{iszero } 0 \longrightarrow \text{true}} \quad \text{E_ISZERO_ZERO} \\
\\
\frac{}{\text{iszero}(\text{succ } v_1) \longrightarrow \text{false}} \quad \text{E_ISZERO_SUCC} \\
\\
\frac{t_1 \longrightarrow t'_1}{\text{iszero } t_1 \longrightarrow \text{iszero } t'_1} \quad \text{E_ISZERO} \\
\\
\frac{}{\text{if true then } t_2 \text{ else } t_3 \longrightarrow t_2} \quad \text{E_IF_TRUE} \\
\\
\frac{}{\text{if false then } t_2 \text{ else } t_3 \longrightarrow t_3} \quad \text{E_IF_FALSE} \\
\\
\frac{t_1 \longrightarrow t'_1}{\text{if } t_1 \text{ then } t_2 \text{ else } t_3 \longrightarrow \text{if } t'_1 \text{ then } t_2 \text{ else } t_3} \quad \text{E_IF}
\end{array}$$

Definition rules: 10 good 0 bad

Definition rule clauses: 14 good 0 bad